INDUSTRONICS ENGINEERING

Industrial Automation & Monitoring Solutions

CONTINUOUS MONITORING SYSTEM

GMP-Compliant Temperature & Humidity Monitoring for Pharma Warehouses

Real-time sensor monitoring • Automated MKT calculation • Email alarms • Role-based dashboard
Audit-trail reporting • 21 CFR Part 11 / EU Annex 11 / GMP & GDP compliant

8	3	4	10 min	5 yrs
Sensor Points	RTU Units	Warehouse Zones	Logging Interval	Data Retention

PRODUCT OVERVIEW

What is the CMS?

The Continuous Monitoring System (CMS) is a purpose-built environmental monitoring platform that continuously tracks Temperature and Relative Humidity across GMP-regulated warehouse areas. Calibrated digital sensors stream live data through a wired network to a centralized, web-based dashboard — giving Quality, Warehouse and Engineering teams real-time visibility, automatic alarms, and audit-ready records, 24 hours a day.

- Continuous, real-time data logging every 10 minutes
- Automatic Mean Kinetic Temperature (MKT) calculation
- Instant email alerts for out-of-limit or sensor faults
- Designed to GMP, GDP, 21 CFR Part 11 & EU Annex 11

DEPLOYED AT

AGP Limited — Plant 01

S.I.T.E. Karachi, Pakistan

Temperature Controlled Approved Area

Raw Material Quarantine Area

Liquid Block Warehouse Area

Secondary Packaging Area

CAPABILITIES

Key Features at a Glance

Live T & RH Monitoring

Continuous real-time readings from calibrated digital sensors at every location.

Automatic MKT

Mean Kinetic Temperature computed automatically per WHO formula, no manual work.

Smart Alarms

Configurable High/Low limits with instant email alerts and malfunction detection.

Trending & Dashboards

Web-based dashboard with live status, historical trends and graphical views.

Role-Based Access

Master, Admin and User roles enforce segregation of duties and accountability.

Data Integrity

Password-protected access, full audit trail, secure on-premise storage.

PDF & Excel Reports

Non-editable PDF reports plus CSV/Excel exports with trend graphs.

24-Hr Auto Recovery

Automatic data recovery for the last 24 hours after any network failure.

HOW IT WORKS

Three-Tier System Architecture



FIELD LAYER

Digital T+RH sensors with local LCD display, RS-485 wiring, deployed at each monitoring point.



COMMUNICATION LAYER

RTU units convert Modbus RTU to TCP/IP and transmit data over a managed Ethernet LAN.



APPLICATION LAYER

On-premise server (PHP + MySQL) hosts the dashboard, alarms, reports and audit trail.

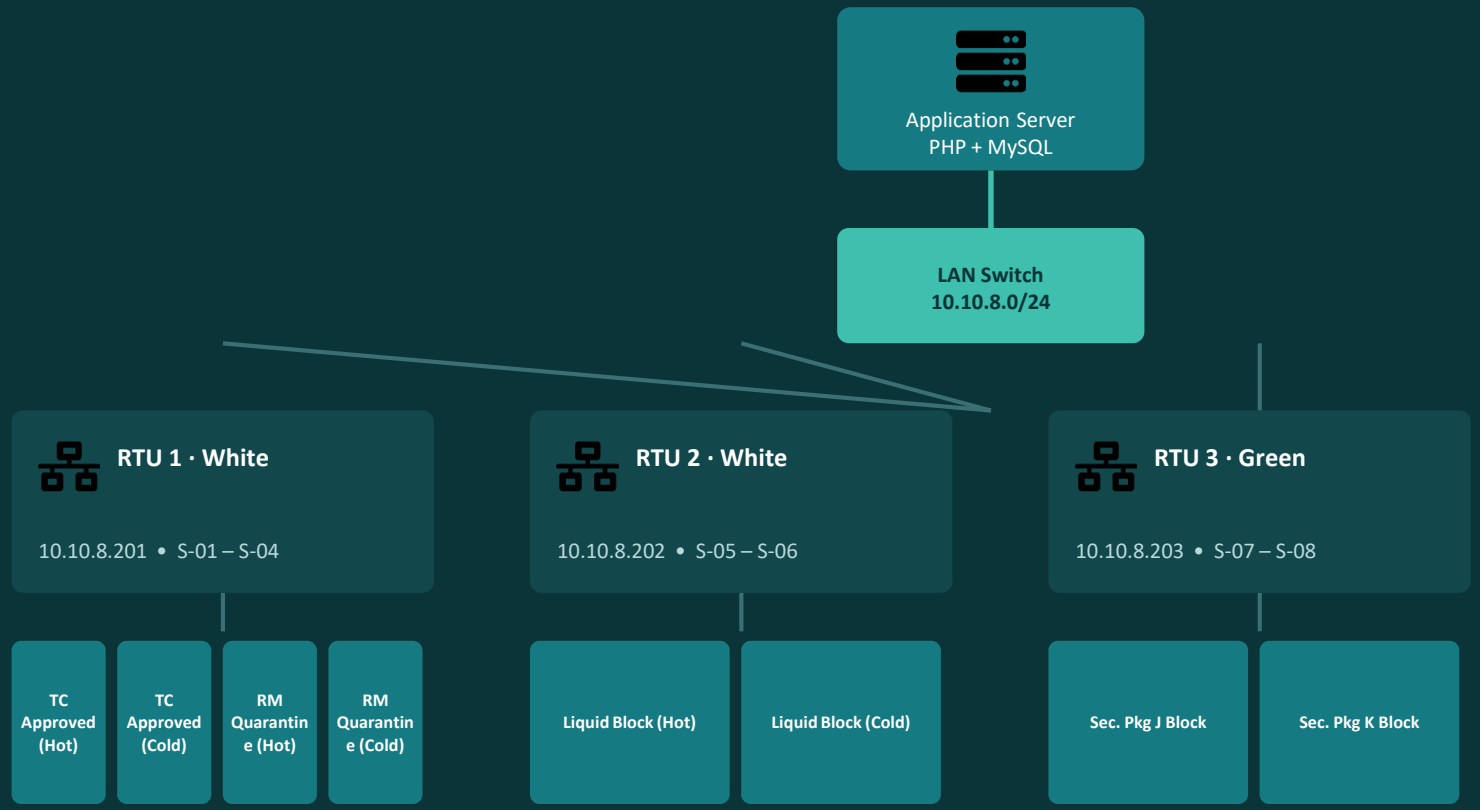
NETWORK & COMMUNICATION SPECIFICATIONS

Sensor → RTU	RTU → Server	Client → Server	Hosting	Compliance
Modbus RTU / RS-485	**Modbus TCP/IP, Port 502**	**HTTP / HTTPS, Port 80/443**	**On-Premise Server**	**GMP • GDP • 21 CFR Part 11**

DEPLOYMENT EXAMPLE

8-Sensor / 3-RTU Reference Deployment

As implemented at AGP Limited Plant 01 — fully scalable to any number of zones and sensors



All sensors measure Temperature (°C) + Relative Humidity (%RH) • RS-485 → Modbus TCP • Static IP per RTU

SOFTWARE

Dashboard & Role-Based Access Control

A single, web-based dashboard — compatible with all major up-to-date browsers — gives every stakeholder the right level of visibility. Live status, historical trends, sensor-wise and area-wise views are available instantly from any LAN-connected device.

- Real-time monitoring status
- Historical data trending & graphs
- Sensor-wise & area-wise visibility
- Works on Chrome, Edge & Firefox

Role	Dashboard	Reports	User Mgmt	All Settings
Master / Admin	✓	✓	✓	✓
Admin	✓	✓	✓	—
User	✓	✓	—	—

Role-Based Access Control (RBAC)

Three-tier RBAC — Master, Admin and User — enforces segregation of duties and ensures every action is attributable, in line with 21 CFR Part 11 and EU GMP Annex 11.



Password-protected login, full audit trail and secure, on-premise data storage protect every record.

RELIABILITY

Alarms, Recovery & Reporting

Alarm System

- Email alerts to concerned personnel
- Configurable High & Low alarm limits
- High/Low temperature alarms
- High/Low humidity alarms
- Sensor & hardware malfunction alarms
- Operator alarm acknowledgement

Network Resilience

If the Ethernet network fails, the CMS automatically recovers and synchronizes the last 24 hours of buffered sensor data once connectivity is restored — no readings are lost.

24 HRS

of buffered data automatically restored after any network malfunction

Reporting Suite

PDF Reports

Non-editable, audit-ready format

Excel / CSV

Exportable for further analysis

Trend Reports

Graphical historical trending

Audit Trail

Mandatory, tamper-evident logging

QUALITY ASSURANCE

Validated for Regulated Industries

The CMS is delivered with a complete validation package — Design, Installation and Operational Qualification (DQ / IQ / OQ) — demonstrating that the system is designed, installed and operating exactly as specified. Every deployment is qualified against the customer's own User Requirement Specification (URS) before go-live.

DQ Design Qualification

Confirms the system design meets every URS requirement before build.

IQ Installation Qualification

Verifies sensors, network, server and software are installed correctly.

OQ Operational Qualification

Functionally tests every feature — alarms, MKT, RBAC, reports — against pass/fail criteria.

STANDARDS & REGULATORY ALIGNMENT

GMP · GDP · 21 CFR Part 11 · EU GMP Annex 11 · WHO Guidelines

WHY INDUSTRONICS ENGINEERING

Engineered for Pharma. Built for Compliance.

End-to-End Delivery

Sensors, network, software, validation and documentation — one accountable vendor.

GMP-First Design

Built from day one around 21 CFR Part 11, EU Annex 11, GMP and GDP requirements.

Full Validation Package

DQ, IQ, OQ, FAT/SAT, calibration certificates and user manuals included.

Scalable Architecture

From a single room to a multi-zone plant, the same platform scales with you.

Ready to bring continuous, compliant monitoring to your facility?

Talk to our engineering team for a site assessment and proposal.

info@industronics.uk

www.industronics.uk